CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. At a party, 15 people each hug every other person exactly once. Number of hugs: 1

(a) 120 **(b)** 112 **(c)** 105 **(d)** 210

2. If $1 + 2 + 3 + \cdots + n = 703$, then n is: 1

(a) 37 **(b)** 36 **(c)** 19 **(d)** 38

3. The sum of the first 30 odd numbers is 900. The sum of the odd numbers from 21 to 59 is: 1

(a) 400 **(b)** 800 **(c)** 779 **(d)** 890

4. The difference between the 20th and the 19th square numbers is: 1

(a) 37 **(b)** 39 **(c)** 41 **(d)** 20

5. A pattern adds the same number every time. Its first term is 7 and its 14th term is 150. The number added each time is: 1

(a) 12 **(b)** 11 **(c)** 10 **(d)** 13

6. A pattern adds 1, then 2, then 3, ... Starting at 10, the term after 12 additions is: 1

(a) 88 **(b)** 154 **(c)** 76 **(d)** 22

7. The sum $1^3 + 2^3 + \cdots + 9^3$ equals: 1

(a) 729 **(b)** 45 **(c)** 90 **(d)** 2025

8. The 41st triangular number minus the 40th triangular number is: 1

(a) 430 **(b)** 40 **(c)** 42 **(d)** 41

9. **Assertion (A):** In the sequence $3, 7, 11, 15, \ldots$ the next term is 19. 1

Reason (R): Each term of this sequence is obtained by adding 4 to the term before it.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The sum of the first 5 odd numbers is 25. 1

Reason (R): 25 is a square number.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. 14 cricket teams play a league in which every team plays every other team once. Find the number of matches. **2**

12. The sum of the first n odd numbers is 961. Find n and the 31st odd number. **2**

13. Find the sum of odd numbers from 19 to 47. **2**

14. 16 cricket teams play a league in which every team plays every other team once. Find the number of matches. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a pattern the 1st term is 14, the 2nd is 23, and so on by adding 9. Which term of the pattern equals 365? Find also the term just before it. **3**

16. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 39. Then find the sum of the odd numbers from 41 to 49. **3**

17. A tower is built in rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 6, and how many in the first 4 rows together? **3**

OR

In a pattern the 1st term is 23, the 2nd is 30, and so on by adding 7. Which term of the pattern equals 331? Find also the term just before it. **3**

18. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 49. Then find the sum of the odd numbers from 51 to 61. **3**

OR

A tower is built in rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 7, and how many in the first 5 rows together? **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ravi makes a row of squares with matchsticks. One square needs 4 sticks, two squares joined side by side need 7 sticks, and three squares need 10 sticks. He keeps adding one square at a time. **5**

(i) How many sticks are needed for 4 squares?

(ii) How many sticks does each new square add?

(iii) How many sticks are needed for 10 squares?

(iv) Ravi has exactly 40 sticks. How many squares can he make in a row? Show how you found it.

20. In a pattern of dot figures, figure 1 has 5 dots and each next figure has 9 more dots than the one before. How many dots are in figure 15? Which figure has 176 dots? **5**

OR

Two sequences: A is $3, 10, 17, 24, \ldots$ and B is $40, 43, 46, 49, \ldots$. Find the 15th term of each and decide which is larger and by how much. **5**

END OF QUESTION PAPER — Total: 40 Marks